

Computer Networking — Chapter 1 Notes

1. What is the Internet?

- **Nuts-and-Bolts View:**
 - The Internet is a global “network of networks,” interconnecting billions of devices worldwide.
 - Devices connected to the Internet are called **hosts** or **end systems** (examples: laptops, smartphones, smart TVs, IoT devices, servers in data centers).
 - These end systems communicate via **communication links** (fiber optics, coaxial cable, wireless radio, satellites) and are interconnected by **packet switches** such as routers and switches.
 - The Internet also includes **Internet Service Providers (ISPs)** which provide connectivity to individuals, businesses, and content providers.
- **Service View:**
 - The Internet is not just physical hardware; it also provides **services to applications**.
 - Examples:
 - Web browsing (HTTP/HTTPS)
 - Email (SMTP, IMAP, POP3)
 - Video streaming (YouTube, Netflix)
 - Social networking (Facebook, Instagram)
 - Messaging (WhatsApp, Signal)
 - Online gaming
 - It acts as a “platform” enabling communication between distributed applications.
- **Why It Matters:**
 - Today’s economy, education, entertainment, healthcare, and government rely on the Internet.
 - Without the Internet, most modern communication, collaboration, and businesses would collapse.

2. What is a Protocol?

- **Definition:**
 - A protocol is a set of rules that define how two or more entities communicate.
 - It specifies:
 1. The **format** of messages.
 2. The **order** in which messages are exchanged.
 3. The **actions** taken when messages are sent and received.
- **Human Analogy:**
 - Imagine a conversation:
 - Person A: “Hi”
 - Person B: “Hello”
 - If Person B doesn’t respond, or says something unexpected, the “protocol” breaks.
- **Computer Examples:**
 - **HTTP (HyperText Transfer Protocol):** Governs how browsers and web servers exchange web pages.

- **TCP (Transmission Control Protocol):** Ensures reliable, ordered delivery of data across the Internet.
 - **IP (Internet Protocol):** Handles addressing and routing of packets.
 - **WiFi Protocols (IEEE 802.11):** Define how devices connect and share a wireless medium.
 - **Why It Matters:**
 - Without protocols, communication would be chaotic.
 - Protocols ensure **predictability**, **compatibility**, and **interoperability** among billions of devices.
 - Protocols are standardized by organizations like the **IETF (Internet Engineering Task Force)**.
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3. Network Edge & Access Networks

- **Network Edge:**
 - End systems sit at the edge of the Internet.
 - These include **clients** (smartphones, laptops) and **servers** (Google, Amazon data centers).
 - Servers often reside in **data centers**, which may contain hundreds of thousands of powerful machines interconnected by high-speed networks.
 - **Access Networks (connecting users to the Internet):**
 - **DSL (Digital Subscriber Line):** Uses existing telephone lines. Typical speeds: 24–52 Mbps download, 3.5–16 Mbps upload.
 - **Cable (Hybrid Fiber Coax):** Uses cable TV infrastructure. Speeds: 40 Mbps–1.2 Gbps download, 30–100 Mbps upload.
 - **Fiber Optics:** Extremely high bandwidth, supports gigabit speeds. Becoming common in cities.
 - **WiFi:** Provides local wireless connectivity within homes, campuses, and offices. Based on IEEE 802.11 standards. Speeds: tens to hundreds of Mbps.
 - **Cellular Networks (4G, 5G):** Provide wide-area wireless access. 5G can achieve 100 Mbps to several Gbps.
 - **Enterprise Networks:** Organizations use Ethernet (1–10 Gbps) and WiFi for connectivity.
 - **Data Center Networks:** Connect massive numbers of servers, often with 10–100 Gbps internal links.
 - **Physical Media:**
 - **Twisted Pair Copper Wire:** Cheap, used in Ethernet and DSL.
 - **Coaxial Cable:** Used in cable Internet.
 - **Fiber Optic Cable:** High speed and long distance.
 - **Wireless Radio:** Includes WiFi, Bluetooth, satellite, and cellular networks.
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4. The Network Core

- **Definition:**
 - The **network core** is the interconnected mesh of routers and high-speed links that carry data packets between end systems.
- **Packet Switching:**
 - Data is divided into small **packets**.
 - Each packet is forwarded independently using the “store-and-forward” principle:
 - A router must receive the entire packet before forwarding it to the next hop.

- Multiple flows share the same link capacity dynamically.
 - **Queueing in Packet Switching:**
 - If many packets arrive at a router faster than they can be sent, they queue up in the router's buffer.
 - If the buffer is full, new packets are dropped (packet loss).
 - **Circuit Switching:**
 - Resources (bandwidth, buffer space) are reserved for a particular call.
 - Example: traditional telephone networks.
 - Multiplexing methods:
 - **FDM (Frequency Division Multiplexing):** Different users get different frequency bands.
 - **TDM (Time Division Multiplexing):** Different users get fixed time slots in rotation.
 - **Comparison:**
 - **Packet Switching:** More efficient for bursty traffic, but delays and losses are possible.
 - **Circuit Switching:** Predictable performance, but inefficient (reserved bandwidth may go unused).
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5. Performance: Delay, Loss, Throughput

- **Types of Delay:**
 1. **Processing Delay:**
 - Time to check packet header and detect errors.
 - Usually in microseconds.
 2. **Queueing Delay:**
 - Time spent waiting in the buffer.
 - Depends on congestion; can range from microseconds to seconds.
 3. **Transmission Delay:**
 - Time required to push all bits of the packet onto the link.
 - Formula: $d_{trans} = L / R$ (L = packet length, R = link bandwidth).
 4. **Propagation Delay:**
 - Time for signals to travel across the medium.
 - Formula: $d_{prop} = \text{distance} / \text{propagation speed}$ ($\sim 2 \times 10^8$ m/s in fiber).
- **Caravan Analogy:**
 - Imagine 10 cars (bits) passing through a toll booth (router).
 - Each car takes 12 seconds to pass (transmission delay).
 - The road trip takes 1 hour (propagation delay).
 - Total delay = transmission + propagation.
- **Traceroute Tool:**
 - Sends special packets with increasing TTL values.
 - Each router along the path sends a reply, allowing you to measure the delay to each hop.
 - Useful for diagnosing network problems and visualizing Internet paths.
- **Packet Loss:**
 - Occurs when packets are dropped due to buffer overflow.
 - Consequence: retransmissions, wasted bandwidth, or reduced application performance.
- **Throughput:**
 - The rate at which bits are successfully delivered to the destination.
 - **Instantaneous throughput:** at a specific time.

- **Average throughput:** over a longer period.
 - **Bottleneck link:** The slowest link in the path determines maximum throughput.
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6. Network Security

- **Original Design:**
 - Internet was built on trust → security was not a priority.
 - Today, security is a major concern.
 - **Common Threats:**
 - **Packet Sniffing:** Capturing unencrypted traffic (like passwords).
 - **IP Spoofing:** Sending packets with a fake source IP.
 - **Denial of Service (DoS):** Flooding a target with bogus traffic.
 - **Distributed DoS (DDoS):** Attack launched from thousands of infected machines (botnets).
 - **Defenses:**
 - **Encryption:** Protects confidentiality (e.g., HTTPS uses TLS).
 - **Authentication:** Ensures identity (e.g., digital certificates).
 - **Integrity Checks:** Prevents tampering (e.g., checksums, digital signatures).
 - **Firewalls and Intrusion Detection Systems:** Block suspicious traffic.
 - **VPNs:** Secure communication over untrusted networks.
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7. Protocol Layers & Models

- **Why Use Layers?**
 - Breaks a complex system into manageable parts.
 - Each layer has a defined role and provides services to the layer above it.
 - Makes updates easier and promotes interoperability.
- **Internet Protocol Stack (5 Layers):**
 1. **Application Layer:** Network applications (HTTP, SMTP, DNS).
 2. **Transport Layer:** Provides process-to-process delivery. (TCP: reliable, UDP: faster but unreliable).
 3. **Network Layer:** Routing and addressing. (IP, routing protocols).
 4. **Link Layer:** Transfers data between directly connected nodes. (Ethernet, WiFi).
 5. **Physical Layer:** Transmission of raw bits over a medium.
- **Encapsulation:**
 - Each layer adds its own header.
 - Message → Segment → Datagram → Frame.
 - Like placing a letter in an envelope, then in a package, then in a box.
- **OSI Model (7 Layers):**
 0. Application
 1. Presentation (encryption, compression, translation)
 2. Session (synchronization, dialog control)
 3. Transport
 4. Network
 5. Data Link

6. Physical

- In practice, Presentation and Session functions are often handled by the application layer itself.
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8. History of the Internet

- **1960s:**
 - Packet switching research.
 - ARPANET (1969): first operational packet-switched network.
 - **1970s:**
 - ALOHAnet (first wireless packet network).
 - Ethernet developed at Xerox PARC.
 - TCP/IP architecture proposed by Cerf & Kahn.
 - **1980s:**
 - Widespread adoption of TCP/IP.
 - DNS introduced.
 - Standardization of email and FTP.
 - **1990s:**
 - ARPANET retired.
 - Tim Berners-Lee invented the World Wide Web.
 - Web browsers (Mosaic, Netscape) popularized Internet use.
 - Commercial ISPs emerged.
 - **2000s:**
 - Broadband and WiFi became widespread.
 - Rise of web applications (Google, Facebook, YouTube).
 - Security became crucial with e-commerce and online banking.
 - **2010s – Present:**
 - Mobile Internet (4G, 5G).
 - Cloud computing (AWS, Azure).
 - Internet of Things (IoT).
 - Software-Defined Networking (SDN).
 - Billions of devices connected globally.
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Summary

- The Internet can be viewed as **infrastructure** (hardware) and as a **service provider** (applications).
 - **Protocols** define how communication happens.
 - Users connect via **access networks**, while the **core** carries data between them.
 - **Performance** is affected by delay, loss, and throughput.
 - **Security** threats exist, but modern defenses mitigate them.
 - The Internet stack (5 layers) vs OSI model (7 layers).
 - The Internet evolved from ARPANET in the 1960s to today's global, mobile, cloud-driven network.
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